

COMMAND NEXUS · CIT 140 · WEEK 1

Problem Brief

Meridian Forge Robotics · The Process Below

LEARNER

Andy Castillo

ARTIFACT

Problem Brief

STATUS

Working document

SECTION 01

Visible Failure

At 2:13am MR-17 accepted Test_K44_START even though it is out of use, after it was accepted the test continued to repeat itself until an operator sent a request to stop it at 2:14am. The request did not stop the tests and returned ACTIVE_INSTRUCTION_PENDING, indicatating that the request was pending and continued running the test.

SECTION 02

Verified Facts

According to the current scheduler, MR-17 has no active test assignment and Test K-44 does not appear in the current test catalog.

According to the incident log, MR-17 accepted TEST_K44_START at 2:13am and the test began repeating shortly after.

According to Elias, TEST_K44_START was in a valid format and came through an internal interface labeled legacy_sync.exe.

SECTION 03

Assumptions and Open Questions

Question: Nia reported that MR-17 was behaving normally before the problem, but were there any issues with MR-17 before 2:13am?

Question: K-44 was retired 186 days ago, but was it actually removed from every copy, service, or schedule?

SECTION 04

System Environment Map

SECTION 04A

People and Roles

Nia is the Night QA Supervisor and has been keeping the incident records. Elias is the Robotics Systems Engineer and has access to the command telemetry. Mara is the Production Operations Manager and manages the production side affected by the problem.

SECTION 04B

Processes and Rules

According to the current scheduler, MR-17 has no active test assignment and K-44 is not in the current test catalog. K-44 was also retired 186 days before the incident and removed from the active workflow.

SECTION 04C

Information and Data

The incident log shows that MR-17 accepted TEST_K44_START at 2:13am and the test began repeating. The stop request returned ACTIVE_INSTRUCTION_PENDING at 2:14am. The current scheduler shows that MR-17 has no active test assignment and K-44 is not in the current catalog. The maintenance record shows that K-44 was retired 186 days ago. The command telemetry shows that TEST_K44_START came through an internal interface labeled legacy_sync.exe.

SECTION 04D

Technology and Tools

According to Elias, the command TEST_K44_START was in a valid format and it came through an internal interface labeled legacy_sync.exe. It's not verified who owns the service or why it is running.

SECTION 04E

Environment and Constraints

Production in the affected QA cell is currently paused according to Mara and needs to be safely running again. According to Nia, MR-17 should not be restarted until the evidence is preserved because restarting it could clear information that is still needed. The full software history also still needs to be checked.

SECTION 05

Investigation Boundary

Inside the Investigation

We will investigate how MR-17 received TEST_K44_START even though K-44 was retired and MR-17 had no active test assignment. We will also investigate what legacy_sync.exe is, who owns it, and why it was running.

Outside for Now

For now we don't need to investigate costs, injuries, or other QA cells because there is no evidence showing that they were affected by the incident.

SECTION 06

Organizational Impact

According to Mara production in the affected QA cell has been paused and needs to be safely running again.

SECTION 07

Problem Statement

MR-17 accepted and continued running TEST_K44_START even though K-44 was retired and MR-17 had no active test assignment. We still don't know why the command was sent or why legacy_sync.exe was running, and production in the affected QA cell has been paused.

SECTION 08

Next Investigation Step

Before restarting we should preserve the evidence we've found so far and further examine what legacy_sync.exe is doing and why it was running.

SHARED ARTIFACT

Working Document Link

[Not entered](#)

WEEK 1 STOPPING RULE

Define before designing.

This Problem Brief records the problem that requires investigation. It does not recommend a solution.